FreshRoute AI

KEEPING EVERY MILE FRESH



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FreshRoute AI

What Is The Problem?

Distributors lack real-time visibility into food freshness during delivery, causing spoilage, waste, and cost.

- Across the food & beverage industry, **freshness loss in transit** is a major but **unseen source of waste**
- Products leave warehouses at ideal temperatures, but **conditions inside trucks go unmonitored**
- Without visibility, distributors rely on static “**best-by” dates and guesswork**, leading to inefficiency and waste
- According to the USDA, **food waste** in the US is estimated at **30-40% of the food supply**



Example: Kung FU Tea x AGI Logistics

Through interviews with warehouse staff and AGI drivers, I found that:

- On average, Kung Fu Tea's NYC distribution team **discards 20–25% of perishable stock monthly** due to freshness uncertainty, not because it's truly spoiled, but because there's no reliable freshness data
- Even small **3–5°F fluctuations** can **cut milk shelf life by up to 40%** and ruin tapioca texture within hours
- This results in an estimated **$1,200–$1,500 in weekly losses** in spoiled milk, fruit, and tapioca pearls
- Temperature inconsistency is worst at **rear compartments of trucks**

This problem **isn't unique** to Kung Fu Tea. **It's a systemic issue for distributors across the food supply chain.** Without live visibility into freshness, distributors are forced to rely on static dates, leading to avoidable waste, financial losses, and reduced product quality by the time ingredients reach stores

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User Pain Points & Interview Insights

AGI Interviewees:

Jason Lim - Warehouse Manager

Maria Sanchez - AGI Delivery Driver

Eric Zhao - Inventory Coordinator

Kevin Tran - Warehouse Associate

Lily Wu - Franchise Owner (Williston Park)



Freshness Data Gap

Major Pain Points

1. **No Real-Time Visibility During Delivery**
    a. Once shipments leave the warehouse, there's no way to track temperature or freshness until arrival
        i. Quote (from Warehouse Manager): "We pack everything perfectly chiled, but once the trucks leave, its a black box. We have no idea what happens after it leaves the warehouse"
2. **Uneven Temperature Zones Inside Trucks**
    a. Different parts of the delivery truck experience different temperatures, especially during long routes or traffic
        i. Quote (from AGI Delivery Driver): "The front of the truck stays cold, but the back heats up fast when we're stuck in traffic"
3. **Inconsistent Quality at Store Arrival**
    a. Stores receive products that vary in freshness and texture, with no explanation
        i. Quote (from Franchisee Owner): "Sometimes the milk lasts all week, other times it spoils in two days. We don't know why but we pay for these products to be fresh"

All stakeholders — warehouse staff, drivers, & store owners — share one core frustration: They can't see or measure real-time freshness during the delivery process.

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Distributors contribute 10-**15%** of **food waste** in the supply chain
(FWRA, 2019)

Over **60%** of **spoilage** occurs during the **last-mile** delivery
(Gartner Supply Chain Research, 2021)

H.T. HACKNEY

McLANE

HARVEST SHERWOOD
FOOD DISTRIBUTORS

Target User

Third Party Distributors

Unlike internal distribution, third-party distributors manage complex, multi-client supply chains with limited operational control. They typically operate refrigerated trucks but with high variability in delivery conditions due to multiple stops, truck loading patterns, and external factors

Existing **freshness monitoring** tech favors **large chains**' in-house operations — Walmart's Wiliot

Third party distributors operate with **net margins** of about **2-3%** in stable years which **food waste** eats into

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Our Solution

Core Features

- Real Time Tracking
- Data-Driven Freshness Score
- Predictive Alerts
- Actionable Suggestions
- Unified Dashboard

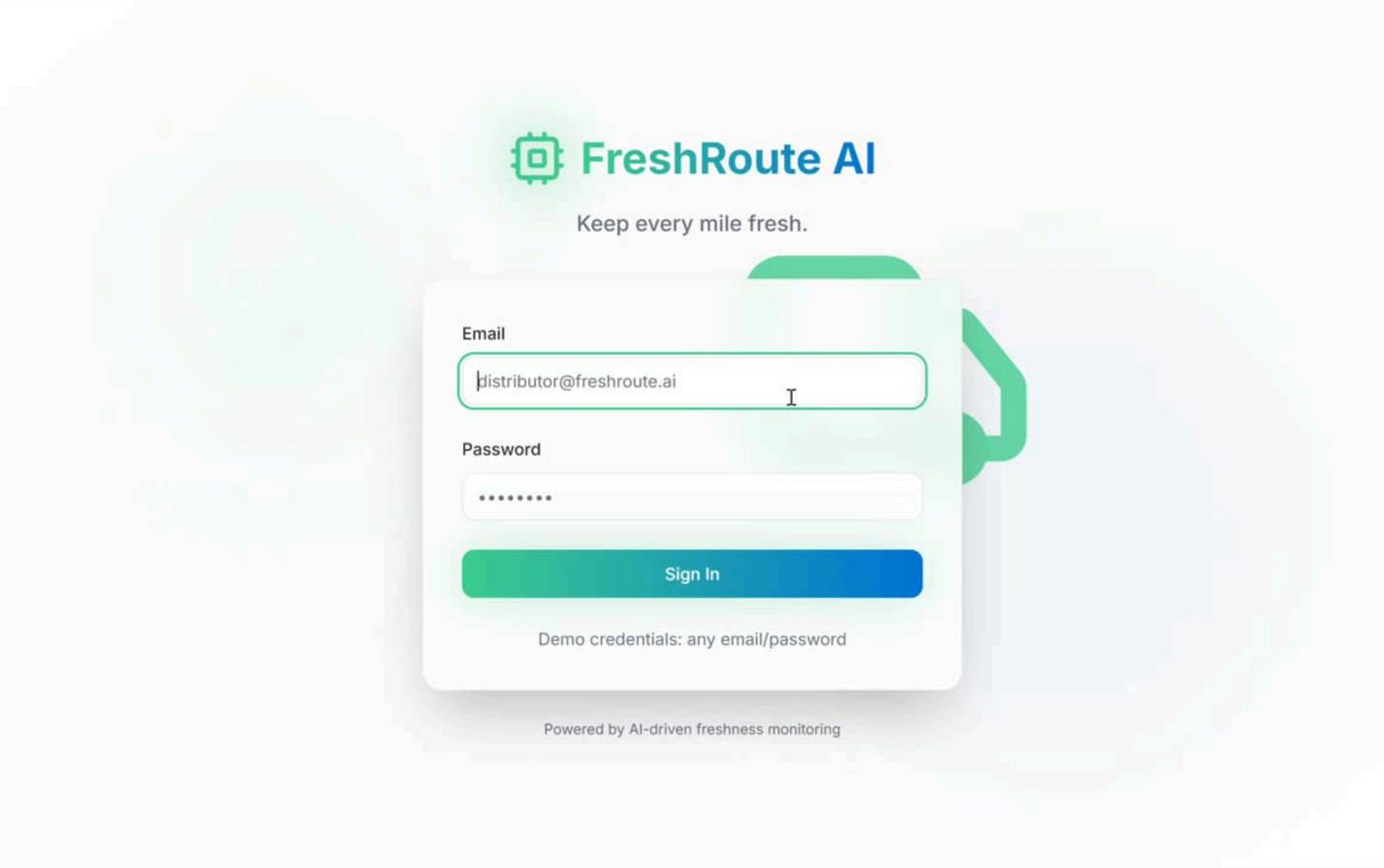


Core Concept

FreshRoute AI is an intelligent **freshness management system** that helps food distributors monitor, predict, and preserve product quality in real time. By combining IoT microchips and AI analytics, it **continuously tracks temperature, humidity, and vibration** across delivery fleets. The platform converts this data into dynamic **Freshness Scores and actionable insights** and even alerting operators before spoilage occurs. Through a unified dashboard, distributors can **visualize every truck and batch**, make faster routing decisions, and reduce food waste across the supply chain.

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Visualizing the Solution

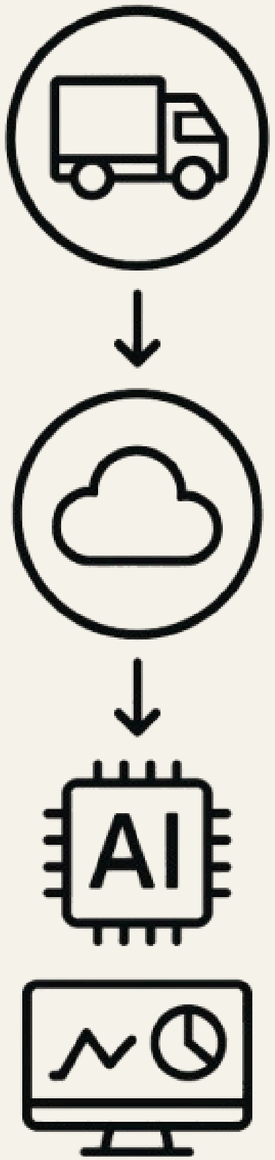


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How it works?

Sensors	Microchips in trucks collect temperature, humidity, and vibrations every 5 minutes
AI Engine	Predicts freshness, detects anomalies, gives recommendations.
Dashboard	Visualizes live fleet conditions with color-coded freshness metrics, trend graphs, and actionable insights for every truck and batch.

FreshRoute AI transforms monitoring into prevention. Instead of reacting to spoilage, distributors can now anticipate issues, act early, and protect product quality. By turning real-time data into timely decisions, the system helps reduce waste, improve efficiency, and ensure that every product reaches retailers in its best condition.



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Tech Specs

Front-End

Requirement	Engineering Notes	Feasibility
Interactive Truck Layout	**React Native**	**Moderate**
Real-Time Temperature Display	**WebSocket** **MQTT** **Redux**	**High**
Freshness Score Indicator	REST API call **React Hooks**	**High**
Alerts & Notifications	**Firebase** Cloud Messaging	**High**
Historical Trend View	**Chart.js** Recharts	**Moderate**

Back-End

Requirement	Engineering Notes	Feasibility
Data Ingestion API	**FastAPI** **MQTT broker Mosquitto**	**High**
Predictive Freshness Model	**Python TensorFlow / scikit-learn**	**Low**
Database Management	**PostgreSQL** + **InfluxDB**	**Moderate**
Notification Engine	**Celery** scheduler **AWS Lambda** event triggers	**High**

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Tech Specs

Cloud Infrastructure

Requirement	Engineering Notes	Feasibility
Scalable Cloud Hosting	**AWS IoT Core Lambda** **API Gateway** **S3**	**High**
Data Storage & Backup	**AWS Timestream S3** backups	**High**
Model Deployment	**AWS SageMaker**	**Low**
Monitoring & Logging	**AWS CloudWatch**	**High**

Hardware

Requirement	Engineering Notes	Feasibility
Temperature & Humidity Sensors	**ESP32** microcontroller **DHT22** sensor	**High**
Data Transmission	**MQTT over LTE Cat M1 / NB-IoT**	**Moderate**
Power Efficiency	**USB-C** power option	**Moderate**
Physical Durability	**Encased PCB**, IP-65 rating	**High**

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Why it works?

Visibility Gap:	Real-time temperature tracking replaces guesswork after trucks leave the warehouse
Uneven Cooling	App shows heat zones, helping optimize product placement inside trucks
Invonsistent Quality:	Distributors verify freshness before hand-off, reducing store complaints

Jason Lin, Warehouse Manager:
“Live temperature tracking would finally let me see issues mid-route, that would cut a lot of waste”

Lily Wu, Franchise Owner (Williston Park):
“Getting freshness data would definitely build our trust and we’d be able to reduce refunduns for spoiled products”

Business Impact & Trust Validation

- **Improves Distributor–Franchisee Trust:** Franchise owners gain transparency into how products are handled.
- **Reduces Product Waste:** AI detection before spoilage means fewer ingredient losses.
- **Increases Revenue:**
  - Fewer refunds or replacements.
  - More consistent drink quality → higher customer satisfaction.
- **Creates a Feedback Loop:** Data collected from each route helps continuously optimize delivery conditions.

Scalability

Cloud-Based B2B SaaS

Scale FreshRoute AI into a cloud-native subscription service, enhancing customer support and integration capabilities. Empower third-party distributors with AI-driven analytics to deliver faster, more accurate freshness insights and actionable recommendations.

Enterprise Supply Chains

Broaden market presence by collaborating with major retailers' in-house distribution networks. Tailor the platform to meet their complex scale, compliance standards, and operational requirements for comprehensive freshness management.

Advanced Machine Learning & NLP Integration

Leverage machine learning and large language models to deliver precise, real-time insights, and edge-case detection plus generating concise, user-friendly summary reports through natural language processing.

Real-Time Shipment Visualization

Enable distributors to monitor all active shipments via an interactive live map, providing transport status, estimated arrival times, and real-time updates for enhanced operational visibility.

Future Features

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Metrics

Reduction in Perishable Waste (%)

- **What It Is:** measures % decrease in spoiled or discarded products
- **Why It Matters:** shows efficiency gains and less financial loss

Average Freshness Score at Delivery

- **What It Is:** Data-based rating of product quality at arrival
- **Why It Matters:** Ensures consistency and customer satisfaction

Cost Savings per Distributor ($/month)

- **What It Is:** tracks money saved from lower waste and better routes
- **Why It Matters:** proves ROI (return on investment) and business impact of FreshRoute AI

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Stakeholders

Engineers

They develop and maintain the data systems and AI models that power FreshRoute AI. Their work ensures the product runs smoothly, delivers accurate insights, and scales effectively as usage grows.

Designers

They create the dashboard and user experience that make complex data easy to understand. Their designs help users act quickly on insights and improve daily operations.

Executives

They guide strategic direction, partnerships, and resource allocation. Their support drives large-scale adoption and long-term business growth.

Users (Distributors)

They directly use FreshRoute AI to monitor freshness and reduce waste, providing real-world feedback that shapes product features. Their adoption determines the product's success and measurable impact.

Thank You